

Лабораторная работа №3

Настройка DHCP-сервера

Гашимова Эсма Эльшан кызы

today

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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Приобретение практических навыков установки и настройки DHCP-сервера Kea на виртуальной машине **server**.

В ходе работы необходимо настроить DHCP-сервер для внутренней виртуальной сети, проверить выдачу IP-адреса клиенту, настроить динамическое обновление DNS-зоны, а также подготовить provisioning-скрипт для Vagrant.

2. 2. Процесс выполнения лабораторной работы

2.1 Установка DHCP-сервера

```
kea-dhcp4.conf  [----] 31 L:[ 25+ 9 34/356] *(1907/16732b) 0093 0x05D

// DHCPv4 configuration starts here. This section will be read by DHCPv4 server
// and will be ignored by other components.
"Dhcp4": {
    // Add names of your network interfaces to listen on.
    "interfaces-config": {
        // See section 8.2.4 for more details. You probably want to add just
        // interface name (e.g. "eth0" or specific IPv4 address on that
        // interface name (e.g. "eth0/192.0.2.1").
        "interfaces": [ "eth1" ]

        // Kea DHCPv4 server by default listens using raw sockets. This ensures
        // all packets, including those sent by directly connected clients
        // that don't have IPv4 address yet, are received. However, if your
        // traffic is always relayed, it is often better to use regular
        // UDP sockets. If you want to do that, uncomment this line:
        // "dhcp-socket-type": "udp"
    },

    // Kea supports control channel, which is a way to receive management
```

Рисунок 1: Установка DHCP-сервера Kea

2.2 Настройка DNS-параметров DHCP

```
[root@server.eegashimova.net eegashimova]#
[root@server.eegashimova.net eegashimova]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-09-15 10:27:15.204 WARN [kea-dhcp4.dhcpsrv/12619.140390549530752] DHCP4_SRV_MT_DISABLED_QUEUE_CONTROL disabling dhcp queue
control when multi-threading is enabled.
2026-09-15 10:27:15.204 WARN [kea-dhcp4.dhcpsrv/12619.140390549530752] DHCP4_RESERVATIONS_LOOKUP_FIRST_ENABLED Multi-threading
is enabled and host reservations lookup is always performed first.
2026-09-15 10:27:15.205 INFO [kea-dhcp4.dhcpsrv/12619.140390549530752] DHCP4_SRV_CFGMGR_NEW_SUBNET4 a new subnet has been added
to configuration: 192.168.1.0/24 with params: t1=900, t2=1800, valid-lifetime=3600
2026-09-15 10:27:15.205 INFO [kea-dhcp4.dhcpsrv/12619.140390549530752] DHCP4_SRV_CFGMGR_SOCKET_TYPE_SELECT using socket type raw
2026-09-15 10:27:15.205 INFO [kea-dhcp4.dhcpsrv/12619.140390549530752] DHCP4_SRV_CFGMGR_ADD_IFACE listening on interface eth1
2026-09-15 10:27:15.205 INFO [kea-dhcp4.dhcpsrv/12619.140390549530752] DHCP4_SRV_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-type"
not specified, using default socket type raw
2026-09-15 10:27:15.205 INFO [kea-dhcp4.dhcpsrv/12619.140390549530752] DHCP4_SRV_LEASE_MGR_BACKENDS_REGISTERED the following lease
backend types are available: memfile
2026-09-15 10:27:15.205 INFO [kea-dhcp4.hosts/12619.140390549530752] HOSTS_BACKENDS_REGISTERED the following host backend types
are available:
2026-09-15 10:27:15.205 INFO [kea-dhcp4.dhcpsrv/12619.140390549530752] DHCP4_SRV_FORENSIC_BACKENDS_REGISTERED the following forensic
backend types are available:
2026-09-15 10:27:15.205 INFO [kea-dhcp4.database/12619.140390549530752] CONFIG_BACKENDS_REGISTERED the following config backend
types are available:
[root@server.eegashimova.net eegashimova]# systemctl --system daemon-reload
[root@server.eegashimova.net eegashimova]# systemctl enable kea-dhcp4.service
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp4.service' → '/usr/lib/systemd/system/kea-dhcp4.service'.
[root@server.eegashimova.net eegashimova]#
```

Рисунок 2: Настройка DNS-сервера и домена в kea-dhcp4.conf

2.3 Настройка DHCP-подсети

```
eegashimova.net  [----] 0 L:[ 1+13 14/ 14] *(244 / 244b) <EOF>
$TTL 1D
@<----->IN SOA<-->@ server.eegashimova.net. (
<-----><-----><-----><-----><----->2026091500<----->; serial
<-----><-----><-----><-----><----->1D<----->; refresh
<-----><-----><-----><-----><----->1H<----->; retry
<-----><-----><-----><-----><----->1W<----->; expire
<-----><-----><-----><-----><----->3H )<-->; minimum
<----->NS<----->@
<----->A<----->192.168.1.1
$ORIGIN eegashimova.net.
server<-->A<----->192.168.1.1
ns<----->A<----->192.168.1.1
dhcp<-->A<----->192.168.1.1
```

Рисунок 3: Настройка сети 192.168.1.0/24 и пула адресов

2.4 Привязка DHCP к интерфейсу

```
192.168.1      [----] 0 L:[ 1+14 15/ 15] *(307 / 307b) <EOF>
$TTL 1D
@<----->IN SOA<-->@ server.eegashimova.net. (
<-----><-----><-----><-----><----->2026091500<----->; serial
<-----><-----><-----><-----><----->1D<----->; refresh
<-----><-----><-----><-----><----->1H<----->; retry
<-----><-----><-----><-----><----->1W<----->; expire
<-----><-----><-----><-----><----->3H )<-->; minimum
<----->NS<----->@
<----->A<----->192.168.1.1
<----->PTR<--->server.eegashimova.net.
$ORIGIN 1.168.192.in-addr.arpa.
1<----->PTR<--->server.eegashimova.net.
1<----->PTR<--->ns.eegashimova.net.
1<----->PTR<--->dhcp.eegashimova.net.
```

Рисунок 4: Привязка DHCP-сервера к интерфейсу eth1

2.5 Проверка конфигурации Kea

```
[root@server.eegashimova.net eegashimova]#  
[root@server.eegashimova.net eegashimova]# systemctl restart named  
[root@server.eegashimova.net eegashimova]# ping dhcp.eegashimova.net  
PING dhcp.eegashimova.net (192.168.1.1) 56(84) bytes of data.  
64 bytes from server.eegashimova.net (192.168.1.1): icmp_seq=1 ttl=64 time=0.028 ms  
64 bytes from server.eegashimova.net (192.168.1.1): icmp_seq=2 ttl=64 time=0.112 ms  
64 bytes from server.eegashimova.net (192.168.1.1): icmp_seq=3 ttl=64 time=0.210 ms  
64 bytes from server.eegashimova.net (192.168.1.1): icmp_seq=4 ttl=64 time=0.080 ms  
64 bytes from server.eegashimova.net (192.168.1.1): icmp_seq=5 ttl=64 time=0.104 ms  
64 bytes from server.eegashimova.net (192.168.1.1): icmp_seq=6 ttl=64 time=0.102 ms  
^C  
--- dhcp.eegashimova.net ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5253ms  
rtt min/avg/max/mdev = 0.028/0.106/0.210/0.054 ms  
[root@server.eegashimova.net eegashimova]#  
[root@server.eegashimova.net eegashimova]# █
```

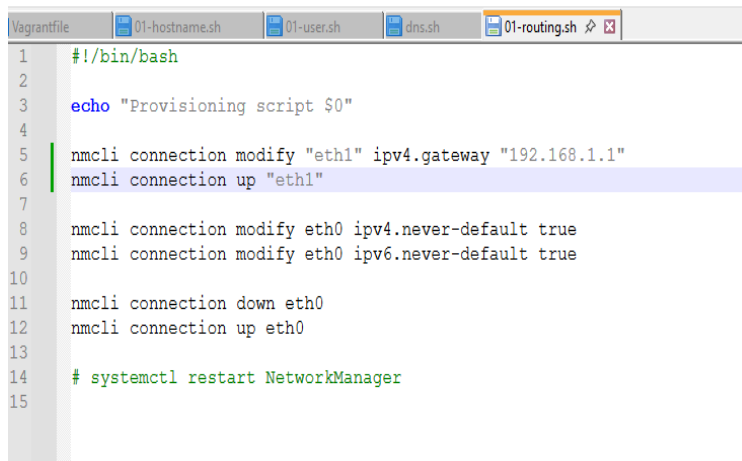
Рисунок 5: Проверка kea-dhcp4.conf и включение службы

2.6 Запись DHCP-сервера в прямой зоне

```
[root@server.eegashimova.net eegashimova]#  
[root@server.eegashimova.net eegashimova]# firewall-cmd --add-service=dhcp  
success  
[root@server.eegashimova.net eegashimova]# firewall-cmd --add-service=dhcp --permanent  
success  
[root@server.eegashimova.net eegashimova]# restorecon -vR /etc  
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0  
[root@server.eegashimova.net eegashimova]# restorecon -vR /var/named/  
[root@server.eegashimova.net eegashimova]# restorecon -vR /var/lib/kea/  
[root@server.eegashimova.net eegashimova]# systemctl start kea-dhcp4.service  
[root@server.eegashimova.net eegashimova]#
```

Рисунок 6: Добавление записи dhcp в прямую DNS-зону

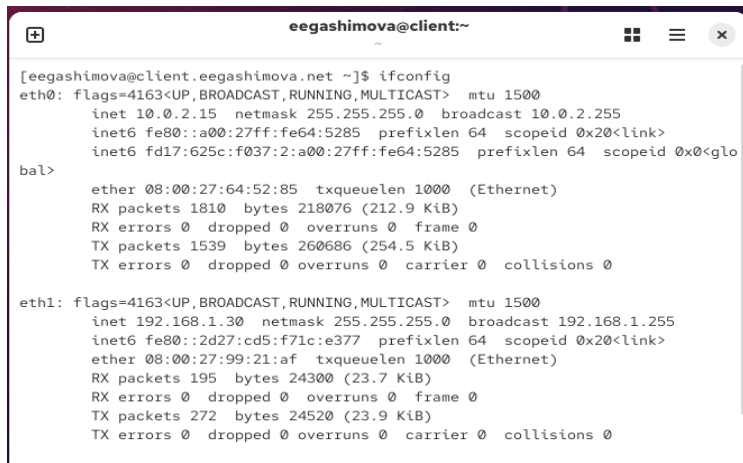
2.7 Запись DHCP-сервера в обратной зоне



```
Vagrantfile  01-hostname.sh  01-user.sh  dns.sh  01-routing.sh ✕
1      #!/bin/bash
2
3      echo "Provisioning script $0"
4
5      nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
6      nmcli connection up "eth1"
7
8      nmcli connection modify eth0 ipv4.never-default true
9      nmcli connection modify eth0 ipv6.never-default true
10
11     nmcli connection down eth0
12     nmcli connection up eth0
13
14     # systemctl restart NetworkManager
15
```

Рисунок 7: Добавление PTR-записи DHCP-сервера

2.8 Проверка имени DHCP-сервера



```
eegashimova@client:~  
[eegashimova@client.eegashimova.net ~]$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255  
    inet6 fe80::a00:27ff:fe64:5285 prefixlen 64 scopeid 0x20<link>  
    inet6 fd17:625c:f037:2:a00:27ff:fe64:5285 prefixlen 64 scopeid 0x0<global>  
    ether 08:00:27:64:52:85 txqueuelen 1000 (Ethernet)  
    RX packets 1810 bytes 218076 (212.9 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 1539 bytes 260686 (254.5 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.1.30 netmask 255.255.255.0 broadcast 192.168.1.255  
    inet6 fe80::2d27:cd5:f71c:e377 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:99:21:af txqueuelen 1000 (Ethernet)  
    RX packets 195 bytes 24300 (23.7 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 272 bytes 24520 (23.9 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рисунок 8: Проверка доступности dhcp.eegashimova.net

2.9 Настройка firewalld и SELinux

```
[root@server.eegashimova.net eegashimova]#  
[root@server.eegashimova.net eegashimova]# cat /var/lib/kea/kea-leases4.csv  
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id  
192.168.1.30,08:00:27:99:21:af,01:00:00:27:99:21:af,3600,1789472140,1,0,0,client,0,,0  
[root@server.eegashimova.net eegashimova]#
```

Рисунок 9: Разрешение DHCP в firewalld и восстановление контекстов SELinux

2.10 Настройка маршрутизации клиента

```
[root@server.eegashimova.net eegashimova]#  
[root@server.eegashimova.net eegashimova]# mkdir -p /etc/named/keys  
[root@server.eegashimova.net eegashimova]# tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key  
[root@server.eegashimova.net eegashimova]# cat /etc/named/keys/dhcp_updater.key  
key "DHCP_UPDATER" {  
    algorithm hmac-sha512;  
    secret "kbvG5deMyW0JhiXmQ/UXFv3m5MH/MPUhSAy2jMiNsG8jaF0vT4yv7Z+F/wcrrk+bstTYMkdYjDDv44FyDf+skA==";  
};  
[root@server.eegashimova.net eegashimova]# chown -R named:named /etc/named/keys/
```

Рисунок 10: Скрипт маршрутизации client через eth1

2.11 Проверка интерфейсов клиента

```
eegashimova.net  [----] 0 L:[ 4+29 33/ 33] *(827 / 827b) <EOF>
//
// ISC BIND named zone configuration for zones recommended by
// RFC 1912 section 4.1 : localhost TLDs and address zones
// and https://tools.ietf.org/html/rfc6303
// (c)2007 R W Franks
//.
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
// Note: empty-zones-enable yes; option is default.
// If private ranges should be forwarded, add
// disable-empty-zone "."; into options
//.

zone "eegashimova.net" IN {
<----->type master;
<----->file "master/fz/eegashimova.net";
<----->update-policy {
<----->    grant DHCP_UPDATER wildcard *.eegashimova.net A DHCID;
<----->};
};

zone "1.168.192.in-addr.arpa" IN {
<----->type master;
<----->file "master/rz/192.168.1";
<----->update-policy {
<----->    grant DHCP_UPDATER wildcard *.1.168.192.in-addr.arpa PTR DHCID;
<----->};
};
```

Рисунок 11: Получение адреса 192.168.1.30 на client

2.12 Проверка DHCP-аренд

```
tsig-keys.json  [----] 2 L:[ 1+ 6 7/ 8] *(185 / 186b) 0010 0x00A
```

```
"tsig-keys": [  
  {  
    <----->"name": "DHCP_UPDATER",  
    <----->"algorithm": "hmac-sha512",  
    <----->"secret": "kbvG5deMyW0JhiXmQ/UXFv3m5MH/MPUhSAy2jMiWsG8jaF0vT4yv7Z+F/wcrrk+bstTYMkdYjDDv44FyDf+sKA=="  
  }  
,  
]
```

Рисунок 12: Содержимое файла kea-leases4.csv

2.13 Создание TSIG-ключа

```
kea-dhcp-ddns.conf [----] 0 L:[ 19+29 48/ 67] *(1649/1926b) 0009 0x009
// See Section 11 for examples and details description.
"DhcpDdns":
{
  "ip-address": "127.0.0.1",
  "port": 53001,
  "control-socket": {
    "socket-type": "unix",
    "socket-name": "kea-ddns-ctrl-socket"
  },
  <?include "/etc/kea/tsig-keys.json"?>
  "forward-ddns" : {
    "ddns-domains" : [
<----->{
<----->   "name": "eegashimova.net.",
<----->   "key-name": "DHCP_UPDATER",
<----->   "dns-servers": [
<-----><----->{ "ip-address": "192.168.1.1" }
<----->   ]
<-----> }
<-----> ]
  },
  "reverse-ddns" : {
    "ddns-domains" : [
<----->{
<----->   "name": "1.168.192.in-addr.arpa.",
<----->   "key-name": "DHCP_UPDATER",
<----->   "dns-servers": [
<-----><----->{ "ip-address": "192.168.1.1" }
<----->   ]
<-----> }
<-----> ]
}
```

Рисунок 13: Генерация ключа DHCP_UPDATER

2.14 Настройка update-policy

```
[root@server.eegashimova.net eegashimova]#  
[root@server.eegashimova.net eegashimova]# chown kea:kea /etc/kea/kea-dhcp-ddns.conf  
[root@server.eegashimova.net eegashimova]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf  
2026-09-15 10:56:46.897 INFO [kea-dhcp-ddns.dctl/17283.139722727324992] DCTL_CONFIG_CHECK_COMPLETE server has completed configuration check: listening on 127.0.0.1, port 53001, using UDP, result: success(0), text=Configuration check successful  
[root@server.eegashimova.net eegashimova]# systemctl enable --now kea-dhcp-ddns.service  
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service' -> '/usr/lib/systemd/system/kea-dhcp-ddns.service'.  
[root@server.eegashimova.net eegashimova]# systemctl status kea-dhcp-ddns.service  
● kea-dhcp-ddns.service - Kea DHCP-DDNS Server  
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disabled)  
   Active: active (running) since Tue 2026-09-15 10:57:01 UTC; 11s ago  
 Invocation: 81b6ffb40fd24e02a58a0c03b2dcbe9  
    Docs: man:kea-dhcp-ddns(8)  
   Main PID: 17492 (kea-dhcp-ddns)  
    Status: "Dispatching packets..."  
     Tasks: 5 (limit: 10233)  
    Memory: 2.3M (peak: 7M)  
       CPU: 11ms  
   CGroup: /system.slice/kea-dhcp-ddns.service  
           └─17492 /usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf
```

Рисунок 14: Разрешение динамического обновления DNS-зон

2.15 Ключ для Кеа

```
// DHCPv4 configuration starts here. This section will be read by dnsmasq server
// and will be ignored by other components.
"Dhcp4": {
  // Add names of your network interfaces to listen on.
  "interfaces-config": {
    // See section 8.2.4 for more details. You probably want to add just
    // interface name (e.g. "eth0" or specific IPv4 address on that
    // interface name (e.g. "eth0/192.0.2.1").
    "interfaces": [ "eth1" ]

    // Kea DHCPv4 server by default listens using raw sockets. This ensures
    // all packets, including those sent by directly connected clients
    // that don't have IPv4 address yet, are received. However, if your
    // traffic is always relayed, it is often better to use regular
    // UDP sockets. If you want to do that, uncomment this line:
    // "dhcp-socket-type": "udp"
  },
  ....
  "dhcp-ddns": {
    <----->"enable-updates": true
  },
  "ddns-qualifying-suffix": "eegashimova.net",
  "ddns-override-client-update": true,
  ....
}
```

Рисунок 15: Файл tsig-keys.json

2.16 Настройка DHCP-DDNS

```
[root@server.eegashimova.net eegashimova]#  
[root@server.eegashimova.net eegashimova]# systemctl restart kea-dhcp4.service  
[root@server.eegashimova.net eegashimova]# systemctl status kea-dhcp4.service  
● kea-dhcp4.service - Kea DHCPv4 Server  
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)  
   Active: active (running) since Tue 2026-09-15 10:59:57 UTC; 11s ago  
 Invocation: 012c06452520456dbacf979bc880ad8c  
    Docs: man:kea-dhcp4(8)  
   Main PID: 17998 (kea-dhcp4)  
    Status: "Dispatching packets..."  
    Tasks: 7 (limit: 10233)  
   Memory: 2.5M (peak: 6.8M)  
      CPU: 13ms  
   CGroup: /system.slice/kea-dhcp4.service  
           └─17998 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf  
  
Sep 15 10:59:57 server.eegashimova.net systemd[1]: Starting kea-dhcp4.service - Kea DHCPv4 Server...  
Sep 15 10:59:57 server.eegashimova.net kea-dhcp4[17998]: 2026-09-15 10:59:57.505 INFO [kea-dhcp4.dhcp4/17998.139636245813376]  
Sep 15 10:59:57 server.eegashimova.net kea-dhcp4[17998]: 2026-09-15 10:59:57.506 INFO [kea-dhcp4.commands/17998.139636245813]  
Sep 15 10:59:57 server.eegashimova.net systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.  
[root@server.eegashimova.net eegashimova]#
```

Рисунок 16: Настройка kea-dhcp-ddns.conf

2.17 Проверка службы DDNS

```
[eegashimova@client.eegashimova.net ~]#  
[eegashimova@client.eegashimova.net ~]$ dig @192.168.1.1 client.eegashimova.net  
  
; <<>> DiG 9.18.33 <<>> @192.168.1.1 client.eegashimova.net  
; (1 server found)  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 13303  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
; COOKIE: 51e05f770f03f93b010000006aa9257febbdae58bded2013 (good)  
;; QUESTION SECTION:  
;client.eegashimova.net.                IN      A  
  
;; ANSWER SECTION:  
client.eegashimova.net. 1200      IN      A      192.168.1.30  
  
;; Query time: 0 msec  
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)  
;; WHEN: Tue Sep 15 11:01:19 UTC 2026  
;; MSG SIZE rcvd: 95  
  
[eegashimova@client.eegashimova.net ~]$ █
```

87 × 26

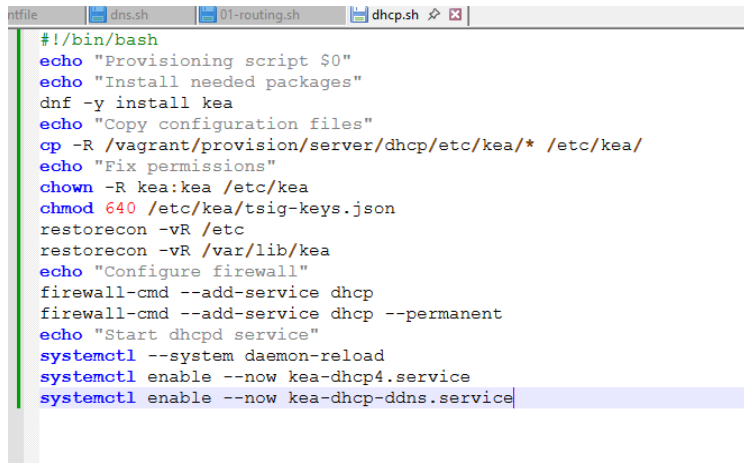
Рисунок 17: Запуск и проверка kea-dhcp-ddns.service

2.18 Включение DDNS в DHCP

```
[root@server.eegashimova.net eegashimova]#  
[root@server.eegashimova.net eegashimova]# cd /vagrant/provision/server/  
[root@server.eegashimova.net server]# mkdir -p /vagrant/provision/server/dhcp/etc/kea  
[root@server.eegashimova.net server]# cp -R /etc/kea/* /vagrant/provision/server/dhcp/etc/kea/  
[root@server.eegashimova.net server]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/  
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y  
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/eegashimova.net'? y  
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol error  
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Protocol error  
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost': Protocol error  
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback': Protocol error  
[root@server.eegashimova.net server]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/  
cp: overwrite '/vagrant/provision/server/dns/etc/named/eegashimova.net'? y  
[root@server.eegashimova.net server]# touch dhcp.sh  
[root@server.eegashimova.net server]# chmod +x dhcp.sh  
[root@server.eegashimova.net server]#
```

Рисунок 18: Включение динамического обновления DNS в kea-dhcp4.conf

2.19 Проверка DHCP-службы



The image shows a terminal window with a title bar containing four tabs: 'ntfile', 'dns.sh', '01-routing.sh', and 'dhcp.sh'. The 'dhcp.sh' tab is active. The terminal displays a series of commands for provisioning a DHCP service. The commands are: `#!/bin/bash`, `echo "Provisioning script $0"`, `echo "Install needed packages"`, `dnf -y install kea`, `echo "Copy configuration files"`, `cp -R /vagrant/provision/server/dhcp/etc/kea/* /etc/kea/`, `echo "Fix permissions"`, `chown -R kea:kea /etc/kea`, `chmod 640 /etc/kea/tsig-keys.json`, `restorecon -vR /etc`, `restorecon -vR /var/lib/kea`, `echo "Configure firewall"`, `firewall-cmd --add-service dhcp`, `firewall-cmd --add-service dhcp --permanent`, `echo "Start dhcpd service"`, `systemctl --system daemon-reload`, `systemctl enable --now kea-dhcp4.service`, and `systemctl enable --now kea-dhcp-ddns.service`. The last two lines are highlighted with a light blue background.

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install kea
echo "Copy configuration files"
cp -R /vagrant/provision/server/dhcp/etc/kea/* /etc/kea/
echo "Fix permissions"
chown -R kea:kea /etc/kea
chmod 640 /etc/kea/tsig-keys.json
restorecon -vR /etc
restorecon -vR /var/lib/kea
echo "Configure firewall"
firewall-cmd --add-service dhcp
firewall-cmd --add-service dhcp --permanent
echo "Start dhcpd service"
systemctl --system daemon-reload
systemctl enable --now kea-dhcp4.service
systemctl enable --now kea-dhcp-ddns.service
```

Рисунок 19: Проверка состояния kea-dhcp4.service

3. 3. Результаты работы



3.1 Настройка DHCP-сервера

На виртуальной машине **server** был установлен DHCP-сервер Kea.

В файле **/etc/kea/kea-dhcp4.conf** была настроена сеть **192.168.1.0/24**.

Для клиентов был задан диапазон адресов **192.168.1.30–192.168.1.199**.

В качестве шлюза и DNS-сервера использовался адрес **192.168.1.1**.

3.2 Проверка работы DHCP

DHCP-сервер был привязан к интерфейсу **eth1**.

После запуска виртуальной машины **client** ей был выдан IP-адрес **192.168.1.30**.

Работа DHCP была проверена с помощью **ifconfig** и файла **/var/lib/kea/kea-leases4.csv**.

3.3 Настройка DDNS

Для динамического обновления DNS был создан TSIG-ключ **DHCP_UPDATER**.

В BIND были настроены правила **update-policy** для прямой зоны **eegashimova.net** и обратной зоны **1.168.192.in-addr.arpa**.

Компонент **kea-dhcp-ddns** был настроен для передачи обновлений DNS-серверу **192.168.1.1**.

3.4 Проверка DDNS

После повторного получения адреса клиентом в DNS автоматически появилась запись:

```
client.eegashimova.net A 192.168.1.30
```

Проверка с помощью **dig** подтвердила корректную работу динамического обновления DNS.

В результате выполнения лабораторной работы был установлен и настроен DHCP-сервер Kea.

Сервер успешно выдаёт IP-адреса клиентам внутренней виртуальной сети.

Также было настроено динамическое обновление DNS-зоны, благодаря чему запись о клиенте автоматически добавляется в DNS после получения адреса по DHCP.

Созданный provisioning-скрипт позволяет автоматически воспроизвести настройку DHCP-сервера при развёртывании виртуальной машины через Vagrant.